Methods memo: usage-based AI exposure across Australia's earnings distribution

Draft for comment (working draft v0.9)

Luc Taya, Economics (Honours) graduate, University of Western Australia

2026-10-26

Purpose. This memo sets out the method of a short descriptive project, with the first two charts, so that reviewers can comment on the method before the working draft is released. The questions at the end are where comments would help most.

What is measured. The usage-based exposure scores come from US task-level Claude.ai use by self-selected users, from one product [1]. No Australian occupational AI use is measured. The scores are applied to Australian occupation mixes, and the project describes how they are spread across Australia's earnings distribution and how they differ from measures of AI potential. It is not a job-loss or career-risk score, and it estimates no effects. DEWR has already converted the same scores to ANZSCO and used them in a model of employment [2]. This project adds earnings, the usage-potential comparison, Western Australia and the split by sex.

Data

Role	Source
Usage-based exposure	Massenkoff and McCrory observed exposure, SOC `2018` (756 occupations, 54% exactly zero)
Potential exposure	JSA automation and augmentation [3]; Eloundou et al. beta [4]; AIOE, aggregate only [5]
Employment	ABS `EQ08`, ANZSCO `v1.2` unit groups, average of May `2025` to February `2026` (14.67 million employed)
Earnings	EEH May `2025`, weekly total cash earnings, all employees; alternatives: EEH hourly (three-digit) and ATO salary or wages
Crosswalks	BLS SOC `2018-2010`, BLS `ISCO-08`-SOC `2010`, ABS ANZSCO-`ISCO-08`; JSA's direct ANZSCO-O*NET correspondence
US weights	BLS OEWS May `2025` employment

Method

From US occupations to Australian units. A unit's usage-based exposure is the weighted mean of the scores of its linked US occupations, with weights renormalised over occupations that have a score. The extensive margin applies the same weights to an indicator of any measured use; intensity is exposure divided by the extensive margin. Four weighting schemes are run throughout: OEWS employment weights (primary), equal weights, title-match-first, and JSA's direct correspondence. Coverage is 99.1% of employment. Each unit is graded by the number of US occupations feeding it; units fed by five or more hold 38.5% of employment, and every result is repeated without them.

Earnings deciles. Units are ranked by EEH weekly earnings and split into employment-weighted deciles, with units that straddle a boundary divided between deciles. Each decile holds exactly a tenth of employment. The project reports decile means, the top-minus-bottom difference in percentage points and the concentration index $2\,\mathrm{cov}(e, r)/\bar{e}$.

Usage against potential. For each unit, the gap is the employment-weighted percentile rank of usage-based exposure minus that of a potential measure, with tied values at the mid-rank, averaged by decile. JSA automation is the primary comparator. The comparison with Eloundou beta is partly mechanical, because the usage-based score is gated on that beta.

What counts as holding. Fixed before results: a decile profile holds if its slope has the same sign as the primary and the Spearman correlation of decile means is at least `0.7`; the gap holds if its sign at both ends is the same under all four schemes. Of 60 specifications, 32 hold.

First charts

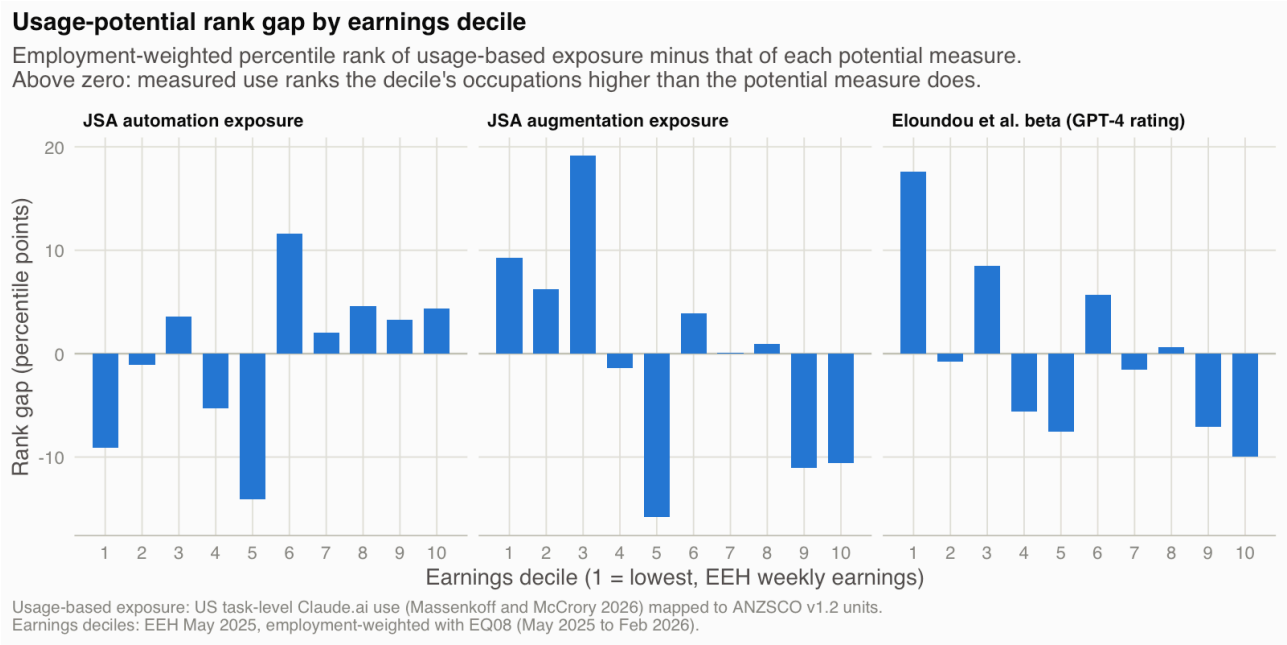


Figure 1: Usage-potential rank gap by earnings decile, main specification. The appendix shows the range across weighting schemes.

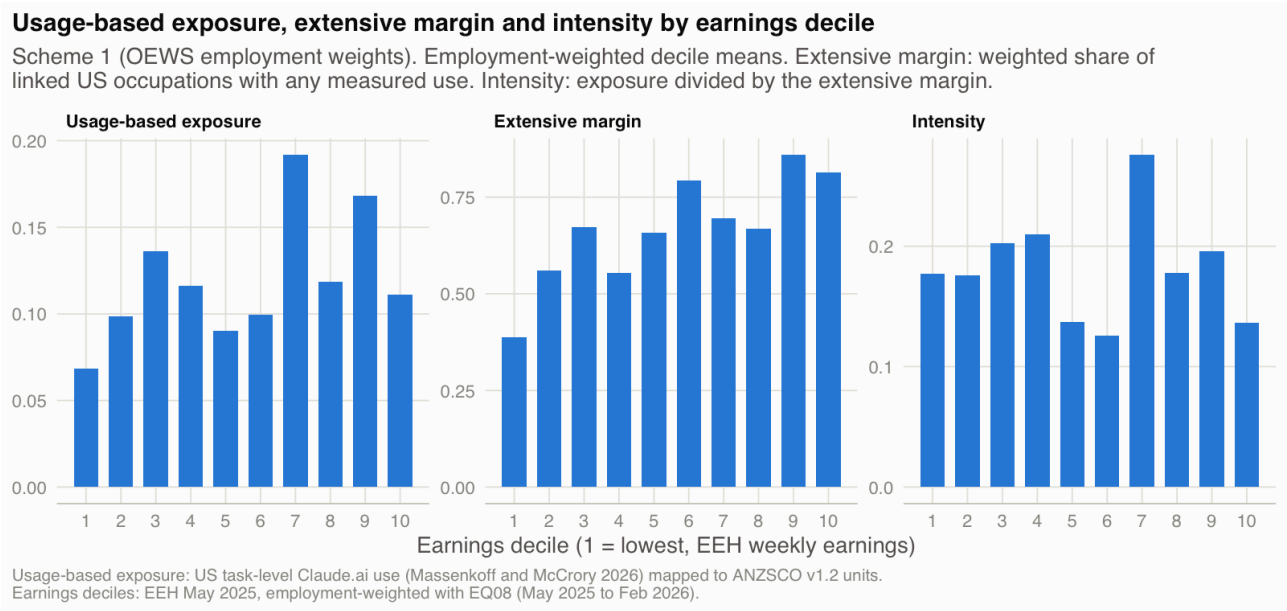


Figure 2: Usage-based exposure, extensive margin and intensity by earnings decile, main specification.

Questions for reviewers

1. **Earnings ranking.** No hourly earnings are published at unit-group level, so the primary ranking uses weekly earnings of all employees, which mixes pay rates with hours. Is that defensible, or should the hourly three-digit series lead?
2. **Crosswalk.** The primary chain runs through `ISCO-08`; JSA's direct correspondence gives a noticeably different profile, largely because of one large unit. Which chain would you trust as primary, and is restricting to non-diffuse units a sensible check?
3. **The gap.** Is a difference in employment-weighted percentile ranks the right way to compare a usage measure with potential measures on different scales? Would you prefer z-scores or levels?
4. **Zeros.** Over half of the scored US occupations are exactly zero because of a usage gate. Should the extensive margin, rather than the score, be the headline?

5. **Independent potential measures.** Beyond JSA and AIOE, is there a measure of AI potential for Australian occupations that is independent of Eloundou et al.'s task ratings?
6. **Related work.** Are you aware of work that reports usage-based exposure by Australian earnings, which a search on 25 October 2026 may have missed?

Contributions and AI use: Luc Taya set the question and makes the judgement calls and interpretation. Claude Code, an AI coding assistant, wrote the pipeline and drafted this memo under Luc's direction.

References

Bibliography

[1] M. Massenkoff and P. McCrory, "Labor market impacts of AI: A new measure and early evidence." [Online]. Available: https://www.anthropic.com/research/labor-market-impacts

[2] Department of Employment and Workplace Relations, "AI and employment in Australia: Monitoring framework and evidence to date," Canberra, Jul. 2026. [Online]. Available: https://www.dewr.gov.au/download/17666/ai-and-employment-australia/43205/ai-and-employment-australia/pdf

[3] Jobs and Skills Australia, "Our Gen AI Transition: Implications for Work and Skills," Canberra, 2025. [Online]. Available: https://www.jobsandskills.gov.au/download/19803/our-gen-ai-transition-implications-work-and-skills/3364/our-gen-ai-transition/pdf

[4] T. Eloundou, S. Manning, P. Mishkin, and D. Rock, "GPTs are GPTs: Labor market impact potential of LLMs," *Science*, vol. 384, pp. 1306–1308, 2024.

[5] E. Felten, M. Raj, and R. Seamans, "Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses," *Strategic Management Journal*, vol. 42, no. 12, pp. 2195–2217, 2021.